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**Hollow House Institute**

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**Government and Military Use of  
Artificial Intelligence:**

**A Governance-First Framework for  
Responsible Deployment**

*A White Paper on Strategic Risk, Oversight, and International Norms*

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# Abstract

Artificial intelligence is rapidly transforming military and national security capabilities — from intelligence analysis and logistics to autonomous systems and cyber operations. This white paper examines the dual imperative facing governments: maintaining strategic advantage while preventing misuse, escalation, and systemic harm. Drawing on a governance-first framework, the paper surveys current and emerging military AI use cases across intelligence, surveillance and reconnaissance, command and control, autonomous systems, and cyber and information operations. It maps the technical, ethical, legal, and geopolitical risk landscape, and proposes a lifecycle governance model encompassing requirements, design, testing, deployment, and decommissioning. The paper articulates core principles — lawfulness, responsibility, reliability, and democratic accountability — and translates them into institutional mechanisms and international norm-building measures. Policy recommendations are directed at national governments, defense ministries, and multilateral bodies. The central argument is that responsible military AI governance is not merely a technical challenge but a constitutional, ethical, and

geopolitical one. States that lead on governance will shape the norms that all actors eventually must live with.

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**Keywords:** *artificial intelligence · military AI · autonomous weapons systems · AI governance · international humanitarian law · responsible AI · national security policy*

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## **1. Executive Summary**

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Artificial intelligence is rapidly transforming military and security capabilities — from intelligence analysis and logistics to autonomous systems and cyber operations. Governments face a dual imperative: maintain strategic advantage while preventing misuse, escalation, and systemic harm.

This white paper outlines:

- Current and emerging military AI use cases across all operational domains
- Key risks spanning technical, ethical, legal, and geopolitical dimensions
- Governance and oversight mechanisms appropriate to the nature and stakes of military AI
- Policy and standards recommendations for responsible, accountable deployment

**Core Argument:** Responsible military AI governance is not merely a technical challenge — it is a constitutional, ethical, and geopolitical imperative. States that lead on governance will shape the norms that all actors eventually must live with.

## 2. Context and Drivers

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The accelerating integration of AI into national security operations is driven by a convergence of strategic, technological, and political forces. Understanding these drivers is essential to designing governance frameworks that are both effective and durable.

- **Strategic competition:** States view AI as a decisive enabler for deterrence, power projection, and decision superiority. The perception that rivals are investing heavily in military AI creates powerful incentives to accelerate domestic programs, often ahead of adequate oversight structures.
- **Data and compute abundance:** Intelligence, surveillance, and reconnaissance (ISR) data, satellite feeds, and cyber telemetry now exceed human analytic capacity by orders of magnitude. AI offers the only scalable means of deriving actionable insight from this volume and velocity of information.
- **Political pressure:** Leaders are incentivized to demonstrate they are not falling behind rivals in the AI competition, creating conditions in which deployment timelines routinely outpace the development of governance frameworks, safety validation, and legal review.

These drivers operate simultaneously and reinforce one another, producing a systemic bias toward faster adoption and a corresponding underinvestment in the institutional infrastructure needed to govern AI responsibly at scale.

### 3. Core Military AI Use Cases

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Military AI applications span the full spectrum of operations, from strategic intelligence to tactical engagement. The following use cases represent the most consequential and widely deployed categories as of 2026, and are the primary subject of governance concern.

#### 3.1 Intelligence, Surveillance, and Reconnaissance (ISR)

AI has become foundational to ISR operations, enabling the processing and analysis of data streams that no human workforce could manage manually. Key applications include:

- **Pattern recognition:** Object detection and classification in satellite imagery, drone video feeds, and synthetic aperture radar — identifying vehicles, fortifications, troop movements, and infrastructure changes at scale.
- **Signal and cyber intelligence:** Anomaly detection within communications traffic, network behavioral analysis, and automated attribution of cyber intrusion activity.
- **Threat prediction:** Predictive models trained on historical and real-time data to forecast force movements, escalation indicators, and adversary decision patterns.



### 3.2 Command, Control, and Decision Support

AI is increasingly embedded in the decision-support layer of military command architectures, with significant implications for the speed and nature of military decision-making:

- **Decision-support systems:** Automated course-of-action (COA) analysis, wargaming simulation, logistics optimization, and resource allocation modelling to support commanders under time pressure.
- **Battlefield awareness:** Real-time fusion of multi-domain sensor data across land, sea, air, cyber, and space domains, presented in unified operational pictures with AI-generated assessments and recommendations.

### 3.3 Autonomous and Semi-Autonomous Systems

The development and deployment of autonomous systems represents the most ethically contested frontier of military AI, engaging fundamental questions of legal compliance and human responsibility:

- Uncrewed aerial, surface, and underwater vehicles employed for reconnaissance, electronic warfare, logistics resupply, and — in some configurations — kinetic strike missions.
- Loitering munitions and collaborative drone swarms with varying and sometimes graduated degrees of autonomous target selection and engagement authority.

3.4 Cyber and Information Operations

- **Offensive capabilities:** Automated vulnerability discovery, exploit generation, and adaptive intrusion systems operating at machine speed.
- **Defensive AI:** Intrusion detection and response, automated patch deployment, and the creation of adaptive deception environments (honeypots, decoy networks) to mislead adversaries.
- **Information operations:** Synthetic media generation, micro-targeted influence campaigns, and counter-disinformation tools designed to detect and degrade adversarial narratives at scale.

| Use Case Domain | Primary Function | Key Governance Concern |
|-----------------|------------------|------------------------|
|                 |                  |                        |

|                    |                                                       |                                                 |
|--------------------|-------------------------------------------------------|-------------------------------------------------|
| ISR                | Data analysis, pattern recognition, threat prediction | Accuracy, bias, over-reliance on AI assessments |
| Command & Control  | Decision support, logistics, battlefield awareness    | Automation bias, speed-accuracy trade-offs      |
| Autonomous Systems | Reconnaissance, strike, electronic warfare            | Human control, IHL compliance, accountability   |
| Cyber & Info Ops   | Intrusion, defense, influence                         | Escalation risk, proportionality, attribution   |

## 4. Risk Landscape

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The risks associated with military AI are multi-dimensional and mutually reinforcing. Effective governance must address each category, recognising that failures in one domain typically compound risks in others.

### 4.1 Technical and Operational Risks

- **Model brittleness and classification error:** AI systems trained on historical or synthetic data may fail unpredictably under novel operational conditions — misclassifying civilians as combatants, friendly vehicles as adversary assets, or benign activity as threatening.

- **Data quality and systemic bias:** Skewed, incomplete, or adversarially manipulated training data can produce AI outputs that are systematically unreliable or discriminatory in ways that are difficult to detect until harm has occurred.
- **Human–machine teaming failures:** Automation bias — the tendency to over-trust AI outputs — is a well-documented failure mode. Conversely, under-trust or poor interface design can cause operators to ignore accurate AI alerts, undermining the value of the system entirely.

## 4.2 Ethical and Legal Risks

- **Accountability gaps:** Existing legal frameworks assign responsibility for unlawful harm to identifiable human actors. When AI systems contribute to lethal decisions across complex kill chains, the diffusion of responsibility creates gaps that current law does not adequately fill.
- **Compliance with international humanitarian law (IHL):** The IHL principles of distinction (between combatants and civilians), proportionality (in collateral harm), and precaution (in attack planning) require human judgement that autonomous systems may be incapable of reliably exercising in complex or ambiguous environments.
- **Dignity and the limits of delegation:** The delegation of life-and-death decisions to machine systems raises profound ethical questions about the moral weight of human life and whether such delegation is compatible with the foundational values of international law and democratic societies.

## 4.3 Strategic and Geopolitical Risks

- **Arms race dynamics:** Competitive pressure to deploy immature or inadequately tested systems to avoid perceived strategic disadvantage creates a race-to-the-bottom on safety and governance that is difficult for any single state to exit unilaterally.
- **Escalation and miscalculation:** The speed and opacity of AI-enabled military systems in crisis scenarios increases the risk of unintended escalation — particularly when adversaries cannot accurately interpret or attribute AI-driven actions.
- **Proliferation and diffusion:** Dual-use AI models and military-origin tools risk diffusion to non-state actors, authoritarian governments, and criminal organisations, extending the harm horizon well beyond the original deployment context.

**Risk Compounding:** Technical failures can trigger legal violations; legal gaps can entrench bad practices; and the absence of international norms allows dangerous technical practices to proliferate. These risk categories are not independent — they amplify one another. Governance frameworks must address all three in concert.

## 5. Governance and Oversight Framework

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A governance-first approach to military AI requires more than policy declarations — it demands institutional architecture, enforceable standards, and a lifecycle approach that embeds oversight from the earliest stages of development through to decommissioning. This section proposes a three-layer framework: guiding principles, institutional mechanisms, and lifecycle controls.

### 5.1 Principles for Military AI

The following principles should serve as the foundational commitments from which all downstream governance requirements are derived:

- **Lawful:** All military AI systems must be designed, tested, and deployed in full compliance with applicable domestic law and international humanitarian law. Legal review must precede deployment, not follow it.
- **Responsible:** Clear and non-diffuse human accountability must be maintained across the entire kill chain or decision chain in which AI operates. Full traceability of AI-influenced decisions must be technically guaranteed.
- **Reliable and Secure:** Systems must be subject to rigorous testing, adversarial red-teaming, and ongoing cyber resilience assessment before and during operational deployment.

- **Aligned with Democratic Values:** Military AI programs must be subject to meaningful parliamentary and congressional oversight, and — where consistent with national security — public transparency about their existence, scope, and safeguards.

## 5.2 Institutional Mechanisms

- **Civilian oversight:** Parliamentary and congressional committees must develop genuine technical expertise in AI — not merely procedural oversight — to provide meaningful scrutiny of defense AI acquisition and deployment decisions.
- **Independent review bodies:** Dedicated AI ethics boards, standing red-team units with access to live systems, and independent safety review panels should be established outside the operational chain of command to preserve impartiality.
- **Interagency coordination:** Defense, foreign affairs, intelligence community, and national standards bodies must operate in concert, with shared frameworks rather than fragmented departmental policies that create governance gaps at the seams.

## 5.3 Lifecycle Controls

Governance cannot be retrofitted onto AI systems after deployment. The following controls must be embedded at each phase of the system lifecycle:



| Lifecycle Phase                 | Required Controls                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Requirements</b>             | Explicit articulation of legal, ethical, and safety constraints as non-negotiable design requirements prior to any development contract award.                                   |
| <b>Design &amp; Development</b> | Threat modelling, safety case documentation, model cards, and formal documentation of known limitations, failure modes, and adversarial vulnerabilities.                         |
| <b>Testing &amp; Evaluation</b> | Scenario-based operational testing, adversarial red-teaming, independent evaluation against IHL compliance criteria, and structured operational piloting before full deployment. |

|                                    |                                                                                                                                                                              |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Deployment &amp; Monitoring</b> | <p>Continuous performance monitoring, mandatory incident reporting, and technically enforced kill-switch and rollback mechanisms operable by authorised human operators.</p> |
| <b>Decommissioning</b>             | <p>Defined and pre-agreed criteria for the retirement of systems found to be unsafe, persistently unreliable, or rendered obsolete by technical or legal developments.</p>   |

## 6. International Norms and Agreements

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No single state can resolve the governance challenges of military AI unilaterally. The systemic nature of the risks — arms race dynamics, escalation, proliferation — requires coordinated international action. This section surveys the existing legal landscape and identifies the most promising avenues for norm development.

### 6.1 Existing Legal Framework

The foundational instruments of international law apply to AI-enabled military systems regardless of whether they were drafted with AI in mind:

- **The UN Charter:** Governs the use of force, including the prohibition of aggression and the right of self-defence — principles that apply equally to AI-enabled kinetic operations.
- **International Humanitarian Law:** The Geneva Conventions and their Additional Protocols impose binding obligations of distinction, proportionality, and precaution that must be met by any weapon system, whether operated by a human or an algorithm.
- **International Human Rights Law:** Applies in all circumstances, imposing requirements of legality, necessity, and proportionality on lethal force even outside armed conflict — including in AI-enabled law enforcement and border security contexts.

## 6.2 Emerging Norms

- **Political declarations:** A growing number of states have issued political declarations on responsible military AI use, including commitments to meaningful human control, legal review, and restraint in autonomous weapons development. These declarations are not legally binding but represent an evolving normative baseline.
- **Lethal Autonomous Weapons Systems (LAWS) at the UN:** Ongoing multilateral discussions under the Convention on Certain Conventional Weapons (CCW) Group of Governmental Experts have produced no binding agreement but have established shared vocabulary and identified key areas of concern, including the meaning and scope of "meaningful human control."

## 6.3 Confidence-Building Measures

In the absence of comprehensive treaty frameworks, confidence-building measures can reduce the risk of miscalculation and lay the groundwork for more durable agreements:

- **Transparency reporting:** States publishing transparency reports on their military AI testing protocols, safety standards, and doctrinal frameworks — without compromising operational security — contribute to a common understanding of acceptable practice.

- **Crisis communication channels:** Dedicated hotlines and incident notification protocols for AI-related military incidents — analogous to the Cold War-era nuclear risk reduction centres — would reduce the risk of misattribution and inadvertent escalation.
- **Joint exercises and verification pilots:** Bilateral or multilateral exercises involving AI-enabled systems, with agreed observation and verification protocols, can build trust and demonstrate good-faith commitments to responsible use among allied states.

## 7. Policy Recommendations

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The following recommendations are directed at three principal audiences: national governments and legislatures; defense ministries and armed forces; and multilateral and international bodies. They are

intended to be actionable within existing institutional structures while building toward more durable long-term governance architectures.

## **7.1 For National Governments**

- Codify a national military AI policy with explicit red lines — for example, a legislatively mandated prohibition on fully autonomous targeting systems operating without meaningful human control over the decision to use lethal force.
- Mandate independent safety and ethics review, conducted by bodies outside the operational chain of command, for all high-risk AI capabilities prior to any operational deployment.
- Require rigorous test, evaluation, verification, and validation (TEVV) against legally defined performance and safety thresholds as a prerequisite for any acquisition approval or deployment authority.
- Invest substantially in AI literacy programs for senior commanders, civilian policymakers, and legislative oversight bodies, ensuring that those responsible for governance possess the technical understanding to exercise it meaningfully.

## **7.2 For Defense Ministries and Armed Forces**

- Adopt comprehensive internal AI governance frameworks, including standardised model cards, AI risk registers, and mandatory incident reporting protocols for all operational AI systems.

- Standardise and formally define human-in-the-loop and human-on-the-loop requirements for specific mission types, in accordance with the nature of the system's outputs and the consequences of its decisions.
- Institutionalise red-teaming and adversarial testing as a standard, resourced, and mandatory component of all AI acquisition and upgrade processes — not an optional or ad hoc activity.
- Integrate fail-safe mechanisms, system-level kill switches, and clear rules of engagement for all AI-enabled systems into acquisition contracts as non-negotiable technical requirements.

### 7.3 For International Cooperation

- Support and actively engage with multilateral forums dedicated to the responsible development and use of military AI, including the CCW Group of Governmental Experts on LAWS and emerging bilateral dialogue frameworks.
- Pursue interoperable technical standards for logging, auditability, and safety verification across allied systems, enabling joint operations while maintaining coherent accountability frameworks.
- Develop shared crisis protocols — agreed in advance — for responding to AI-related military incidents, including misfires, autonomous system malfunctions, and misattribution events, to reduce the risk of escalation under conditions of uncertainty.

**Priority Action:** The single highest-leverage action available to most governments is the establishment of a mandatory, independent AI safety and ethics review process as a gating condition for any high-risk military AI capability entering operational service. No deployment without review should become a firm institutional norm, not an aspiration.

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## **8. Conclusion**

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Military AI is not optional for states — it is already embedded in intelligence processing, logistics management, and cyber operations worldwide. The question before governments, armed forces, legislatures, and international bodies is not whether to engage with military AI, but how to govern it in ways that are lawful, accountable, and consistent with the values of democratic societies and the rules of the international order.

A governance-first approach is not a counsel of technological timidity. It does not require states to forfeit strategic advantage or abandon capability development. What it does require is that the institutional infrastructure of governance — legal review, independent oversight, safety testing, accountability



mechanisms, and international norms — be developed in parallel with, and as a condition of, the capabilities themselves.

The risks of the alternative are not hypothetical. AI systems that misclassify targets, operate without meaningful human oversight, or accelerate crises beyond the pace of human decision-making represent strategic liabilities, not just ethical failures. States that deploy AI recklessly invite the kind of incidents that erode domestic trust, damage alliances, and invite international condemnation.

Responsible military AI is not merely a technical challenge; it is a constitutional, ethical, and geopolitical one. States that lead on governance will shape the norms that everyone else will eventually have to live with. The window for establishing durable, legitimate, and broadly accepted frameworks remains open — but it is narrowing with each year of uncoordinated deployment and deferred decision-making.

The Hollow House Institute calls on governments, defense ministries, legislatures, and international bodies to treat AI governance not as a constraint on military capability but as the condition under which such capability can be lawfully, safely, and sustainably exercised. The moment for principled leadership is now.

**Central Finding:** States that invest in governance infrastructure alongside capability development will achieve more durable strategic advantage than those that race to deploy without oversight — because trust, legality, and international legitimacy are themselves strategic assets.

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*Independent Research for Governance, Technology, and Public Interest*

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